Zackary Weissman-Bennett

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SUMMARY

Mechanical Engineering senior at UMass Amherst with experience in systems modeling, multidisciplinary hardware integration, FEA, and technical research. Interested in systems engineering, modeling and simulation, R&D, and solving complex multidisciplinary engineering problems.

EDUCATION

University of Massachusetts - Amherst **Anticipated: May 2027**

Bachelor of Science in Mechanical Engineering **Major GPA: 3.7**

Awards: Dean's List: Fall '25, Spring '25, Spring '26, John & Abigail Adams MCAS Scholarship, Somerville Municipal Scholarship

COURSES

Design of Mechanical Components & Assemblies, Manufacturing Processes, Additive Manufacturing, Mechanics of Materials, Advanced Fluid Mechanics, Thermodynamics, Heat Transfer, System Dynamics, Intro to Python, Mechanical Engineering Lab.

RESEARCH & TECHNICAL EXPERIENCE

UMass CubeSat - Mechanical Engineering team June 2026 - Present

- Develop a 3D CAD model of a 6U CubeSat chassis and evaluate ADCS integration, including placement and packaging of the reaction wheel, motor, flight computer, and IMU.
- Define component envelope, mounting, and additive-manufacturing requirements to support development of a 3D-printed chassis and ADCS hardware.

Ocean Renewable Energy Lab - Undergraduate Research Intern Sep 2025 - Present

Ansys Structural Analysis & Optimization of OTEC Cold Water Pipe with Stiffeners

- Conducted static and eigenvalue buckling analyses of a 3-pipe OTEC system under hydrostatic & internal Suction loading to evaluate collapse risk & determine optimal material, dimensions, and stiffener spacing.
- Developing and refining FEA models in ANSYS, defining boundary conditions and external pressure distributions to simulate operating conditions and assess structural stability margins.

Senior Capstone - MathWorks Electric Motor Fault Detection — Simulation & Modeling Lead Fall 2026 - Present

- Lead development of a MATLAB/Simulink model of a three-phase electric motor, simulating healthy and fault conditions to generate labeled data for machine-learning-based fault detection.
- Collaborate with signal-processing, machine-learning, and hardware team members to define simulation outputs and support comparison with data collected from physical motor testing.

CNC Machining of Standoff Mount - Work Group Leader Fall 2025

- Led a work group within a 30-person engineering team to design and CNC-machine a PCB standoff mount for an assembly targeted for 10,000 units/year production.
- Applied DFMA principles and coordinated with project managers to improve manufacturability, reducing machining setup time by 15%.

OTHER EXPERIENCE

Undergraduate Teaching Assistant - Thermodynamics & Aerospace Fluid Mechanics Aug 2025 - May 2026

- Hosted exam reviews for 50+ students and weekly office hours supporting 15+ students with coursework.
- Managed course communications & organized heating plant tours. Graded exams, homework, and projects.

SKILLS

Design & Analysis: SolidWorks (CAD), GD&T, ANSYS (FEA), XFOIL, by-hand stress analysis, QBlade

Manufacturing: CNC Machining, Manual Milling, 3D Printing, Mechanical Assembly & Testing, Additive Manufacturing

Programming & Tools: MATLAB, Python, Simulink, Machine Learning, Excel